PROJECT PORTFOLIO

RFM Segmentation Analysis

Automated RFM Modeling & Customer Segmentation using Python

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Business Intelligence at MSME Laundry Services

SWIPE <<<

INTRODUCTION

Project Overview

INDUSTRY:
MSME Laundry Services

Fast-growing micro, small, and medium-sized enterprise (MSME) industry in Indonesia, driven by urban mobility, busy lifestyles, and the need to outsource daily household chores.

ROLE:
Business Intelligence

Turn daily operational data from September 2025 to February 2026, spanning late-year activity (Sep–Dec 2025) and early-year trends (Jan–Feb 2026), into actionable strategies that increase customer lifetime value, optimize machine usage, and protect margins.

OBJECTIVE:
Implement Cohort Analysis

To transform raw transactional records into an automated RFM segmentation pipeline, categorizing the customer base into actionable behavioral segments to optimize targeted retention strategies and maximize customer lifetime value.

CONCEPT DEVELOPMENT

Methodology

Raw Transaction Data

Clean raw transaction data and aggregate records at the customer level to compute order frequency, lifetime spending, and days since last purchase.

Create RFM Score

Standardize customer behavior by binning recency, frequency, and monetary metrics into relative quintile scores from 1 to 5.

Create RFM Segmentation

Combine individual scores into unified 3-digit identifiers and apply regex mapping rules to group customers into actionable business segments.

Visualization

Build interactive visual treemaps to map segment proportions and evaluate overall customer base composition.

Key Findings & Business Implications

Technical Implementation

Raw Transaction Data

The raw transaction dataset is imported and filtered for essential columns (Date, Order ID, Customer ID, Sales), then aggregated using Pandas to compute each customer's unique order count, latest order date, and total sales value. To operationalize recency, a reference benchmark (today) is established from the maximum dataset date, allowing the pipeline to calculate d_sinlas (days since last sale) as the baseline for behavioral analysis.

```python
import pandas as pd
import numpy as np
import datetime as dt
```

Python

```python
# 1. importing data
df = pd.read_excel(r'C:\Backup_D\Project\rekap_transaksi.xlsx', sheet_name=4)
df.head()

# 2. choose primary column
df2 = df[['Date', 'Order ID', 'Customer ID', 'Sales']].copy()
df2.head()

# 3. create rfm calculation
df_user = (
    df2.groupby('Customer ID', as_index=False)
    .agg(
        order_cnt=('Order ID', 'nunique'),
        max_order_date=('Date', 'max'),
        total_order_value=('Sales', 'sum')
    )
)

df_user.head()

# 4. create d_sinlas
today = df2['Date'].max()
df_user['d_sinlas'] = (today - df_user['max_order_date']).dt.days

df_user.head()
```

Python

```python
# 5. recency score
df_user['recency_score'] = (
    pd.cut(
        df_user['d_sinlas'],
        bins=[
               df_user['d_sinlas'].min(),
               np.percentile(df_user['d_sinlas'], 20),
               np.percentile(df_user['d_sinlas'], 40),
               np.percentile(df_user['d_sinlas'], 60),
               np.percentile(df_user['d_sinlas'], 80),
               df_user['d_sinlas'].max()
        ],
        labels=[5,4,3,2,1],
        include_lowest=True
    )
    .astype(int)
)

df_user.head()
```

[ ] Python

```python
# 6. frequency score
df_user['frequency_score'] = (
    pd.cut(
        df_user['order_cnt'],
        bins=[
               0,
               np.percentile(df_user['order_cnt'], 20),
               np.percentile(df_user['order_cnt'], 40),
               np.percentile(df_user['order_cnt'], 60),
               np.percentile(df_user['order_cnt'], 80),
               df_user['order_cnt'].max()
        ],
        labels=[1,2,3,4,5],
        include_lowest=True
    )
    .astype(int)
)

df_user.head()
```

[ ] Python

```python
# 7. monetary score
df_user['monetary_score'] = (
    pd.cut(
        df_user['total_order_value'],
        bins=[
               df_user['total_order_value'].min(),
               np.percentile(df_user['total_order_value'], 20),
               np.percentile(df_user['total_order_value'], 40),
               np.percentile(df_user['total_order_value'], 60),
               np.percentile(df_user['total_order_value'], 80),
               df_user['total_order_value'].max()
        ],
        labels=[1,2,3,4,5],
        include_lowest=True
    )
    .astype(int)
)

df_user.head()
```

[ ] Python

Create RFM Score

Customer metrics are translated into standardized scores from 1 to 5 using quintile-based binning (pd.cut with percentile thresholds), where Recency assigns higher scores to recent buyers, and Frequency and Monetary dimensions rank customers based on order volume and spending. This relative ranking normalizes the behavioral dimensions against the specific purchasing patterns of the laundry business.

Create RFM Segmentation

Individual recency, frequency, and monetary scores are concatenated into a three-digit identifier (RFM_score_segment) that captures each customer's comprehensive transaction profile. A regular expression mapping dictionary then translates these combinations into actionable business segments—such as Champions, Loyal Customers, and At Risk—enabling targeted retention and marketing strategies.

```python
# 8. create new df
df_user2 = (
    df_user[[
        'Customer ID',
        'recency_score',
        'frequency_score',
        'monetary_score'
    ]]
    .copy()
)
df_user2.head()

# 9. create rfm score segment
df_user2['RFM_score_segment'] = (
    df_user2['recency_score'].astype(str)
    +
    df_user2['frequency_score'].astype(str)
    +
    df_user2['monetary_score'].astype(str)
)

df_user2.head()
```

Python

```python
# 10. segment mapping
segment_mapping = {
    r'[1-2][1-2][1-5]': 'Hibernating',
    r'[1-2][3-4][1-5]': 'At Risk',
    r'[1-2][5][1-5]': 'Can\'t Loose',
    r'[3][1-2][1-5]': 'About to Sleep',
    r'[3][3][1-5]': 'Need Attention',
    r'[3-4][4-5][1-5]': 'Loyal Customers',
    r'[4][1][1-5]': 'Promising',
    r'[5][1][1-5]': 'New Customers',
    r'[4-5][2-3][1-5]': 'Potential Loyalists',
    r'[5][4-5][1-5]': 'Champions'
}

df_user2['Segment'] = (
    df_user2['RFM_score_segment']
    .replace(segment_mapping, regex=True)
)

df_user2.head()
```

Python

```python
import plotly.express as px

df_grouped = df_user2.groupby('Segment').size().reset_index(name='customer_count')

fig = px.treemap(
    df_grouped,
    path=['Segment'],
    values='customer_count',
    title='<b>RFM Customer Segments Distribution</b>',
    color='customer_count',
    color_continuous_scale='Tealgrn',
)

fig.update_traces(
    textinfo='label+value+percent root',
    texttemplate='<b>%{label}</b><br>Count: %{value:,}<br>Share: %{percentRoot:.1%}',
    textposition='middle center',
)

fig.update_layout(
    margin=dict(t=50, l=25, r=25, b=25),
    title_font=dict(size=18, family='Arial'),
)

fig.show()
```

Python

Visualization

The segmented customer data is aggregated and visualized using an interactive Plotly Treemap (px.treemap) with a professional teal-green gradient (Tealgrn) scaled by customer volume. Custom text templates display segment labels, precise customer counts, and percentage shares relative to the total base, giving stakeholders an immediate, scannable overview of clientele distribution.

OUTPUT SYNTAX

RESULT

	Date	Order ID	Customer ID	Weight	Piece	Piece Type	Service Speed	Service	Sales	Region
0	2025-09-01	1001	INE729	0.96	NaN	NaN	Reguler	Complete Wash	24.00	GJM
1	2025-09-01	1002	PER783	3.38	NaN	NaN	Reguler	Complete Wash	27.04	GJM
2	2025-09-01	1002	PER783	0.00	Small Bed Sheet Set	Bed Sheet	Piece	Piece	15.00	GJM
3	2025-09-01	1003	PUR183	4.49	NaN	NaN	Reguler	Complete Wash	35.92	GJM
4	2025-09-01	1003	PUR183	0.00	Small Bed Sheet Set	Bed Sheet	Piece	Piece	15.00	GJM

1. importing data

	Date	Order ID	Customer ID	Sales
0	2025-09-01	1001	INE729	24.00
1	2025-09-01	1002	PER783	27.04
2	2025-09-01	1002	PER783	15.00
3	2025-09-01	1003	PUR183	35.92
4	2025-09-01	1003	PUR183	15.00

2. choose primary column

	Customer ID	order_cnt	max_order_date	total_order_value
0	ADE488	4	2026-01-17	136.16
1	ADH374	30	2026-02-22	1107.28
2	ADH375	5	2026-02-04	216.24
3	ADI693	5	2026-02-16	143.36
4	ADJ966	5	2025-12-29	224.76

3. create rfm calculation

	Customer ID	order_cnt	max_order_date	total_order_value	d_sinlas
0	ADE488	4	2026-01-17	136.16	42
1	ADH374	30	2026-02-22	1107.28	6
2	ADH375	5	2026-02-04	216.24	24
3	ADI693	5	2026-02-16	143.36	12
4	ADJ966	5	2025-12-29	224.76	61

4. create d_sinlas

	Customer ID	order_cnt	max_order_date	total_order_value	d_sinlas	recency_score	frequency_score	monetary_score
0	ADE488	4	2026-01-17	136.16	42	1	1	1
1	ADH374	30	2026-02-22	1107.28	6	4	4	4
2	ADH375	5	2026-02-04	216.24	24	2	1	2
3	ADI693	5	2026-02-16	143.36	12	3	1	1
4	ADJ966	5	2025-12-29	224.76	61	1	1	2

5,6,7. create recency, frequency, and monetary score

	Customer ID	recency_score	frequency_score	monetary_score
0	ADE488	1	1	1
1	ADH374	4	4	4
2	ADH375	2	1	2
3	ADI693	3	1	1
4	ADJ966	1	1	2

8. create new df

	Customer ID	recency_score	frequency_score	monetary_score	RFM_score_segment
0	ADE488	1	1	1	111
1	ADH374	4	4	4	444
2	ADH375	2	1	2	212
3	ADI693	3	1	1	311
4	ADJ966	1	1	2	112

9. create rfm score segment

	Customer ID	recency_score	frequency_score	monetary_score	RFM_score_segment	Segment
0	ADE488	1	1	1	111	Hibernating
1	ADH374	4	4	4	444	Loyal Customers
2	ADH375	2	1	2	212	Hibernating
3	ADI693	3	1	1	311	About to Sleep
4	ADJ966	1	1	2	112	Hibernating

10. segment mapping



11. visualization

IMPACT

Key Findings & Business Implications

The Insight

The RFM segmentation reveals that 49.2% of customers (221 out of 449) belong to declining or inactive groups (Hibernating, At Risk, About to Sleep, and Need Attention), indicating weak repeat purchasing. Conversely, Champions and Loyal Customers make up 32.3%, showing a solid core of repeat buyers. Given the business's proximity to a university campus, this high inactivity likely reflects seasonal demand drops during academic holidays.

The Business Implication

The business should prioritize reactivation and retention strategies. Targeted efforts should focus on At Risk customers who previously bought frequently, alongside seasonal promotions for Hibernating users as the university semester resumes. Simultaneously, strengthening loyalty programs for Champions and Loyal Customers will secure recurring revenue and reduce heavy reliance on new customer acquisition.

Thank You!

Note:
All data has been modified and anonymized to protect business confidentiality, while the analysis process remains representative of the actual project.

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